國立臺北科技大學
National Taipei University of Technology

資料結構

Data Structure

Lab 03

姓名：簡紘田

學號：114310115

Q1:Leetcode 1394

Briefly describe the problem. (題目簡述)

Given an array of integers arr, a lucky integer is an integer that has a frequency in the array equal to its value.

Return the largest lucky integer in the array. If there is no lucky integer return -1.

Code

```
class Solution {        // 定義一個名為 Solution 的類別
public:
    int findLucky(vector<int>& arr) {
        int count[101] = { 0 }; // 建立一個大小 101 的陣列，並將全部數值設為 0，用來計
                                   數字出現的次數

        for (int num : arr) {   // 依序將 arr 中的資料取出命名為 num
            count[num]++;   // 將與取出數字相對應位置的數字 +1，紀錄為出現過一次
        }

        for (int i = 100; i >= 1; i--) { // 從一百倒著數到一，一一尋找最大的 Lucky number
            if (count[i] == i){     // 數字出現的次數是否與本身相等
                return i;           // 如果是則回傳此 Lucky number
            }
        }

        return -1;              // 如果找不到 Lucky number 則回傳 -1
    }
};
```

Result

Problem List | Submit | Premium

Description | Editorial | Solutions | Submissions

1394. Find Lucky Integer in an Array

Easy | Topics | Companies | Hint

Given an array of integers `arr`, a **lucky integer** is an integer that has a frequency in the array equal to its value.

Return *the largest* ***lucky integer*** *in the array*. If there is no **lucky integer** return `-1`.

Example 1:

```
Input: arr = [2,2,3,4]
Output: 2
Explanation: The only lucky number in the array is 2 because frequency[2] == 2.
```

Example 2:

```
Input: arr = [1,2,2,3,3,3]
Output: 3
Explanation: 1, 2 and 3 are all lucky numbers, return the largest of them.
```

Example 3:

```
Input: arr = [2,2,2,3,3]
Output: -1
Explanation: There are no lucky numbers in the array.
```

Constraints:

- `1 <= arr.length <= 500`
- `1 <= arr[i] <= 500`

Seen this question in a real interview before? 1/5

Yes | No

Accepted 334,940/443.5K | Acceptance Rate 75.5%

Topics

Companies

1.6K | 176 | 12 Online

Code

C++ | Auto

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        }

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```

Saved | Ln 18, Col 3

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

arr =
[2,2,3,4]

Output

2

Expected

2

Contribute a testcase

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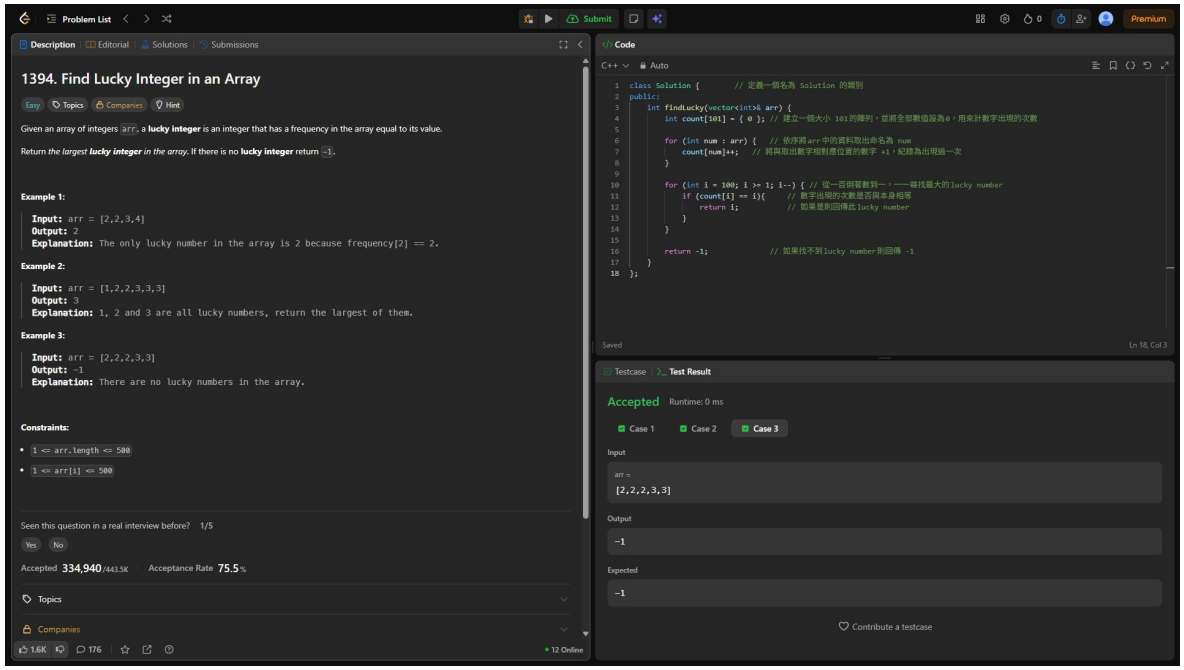

Discussion

Total operations:

101 (initialization)

+ 2n (first loop operations)

+ 400 (second loop operations)

+ 1 (return)

= 101 + 2n + 400 + 1

= 2n + 502 operations

Therefore, O(n) complexity